Assignment-2 DBMS

Name: Vanshika Dubey
Roll No.: 2401010067

Section-A

1- Fill in the blanks

1- The ER Diagram visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.

2- In the Relational Model, data is organized into tables, with relationships between entities defined through foreign key constraints.

3- The Object-Oriented Model treats data as a collection of objects, each with unique attributes and methods, mirroring real-world entities and interactions.

4- Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information.

Section - B (case study based)

2- In a healthcare system, discuss how data abstraction can simplify interaction between healthcare professionals and patient records.

Data abstraction is an important concept in database systems that hides complex data structures and shows only necessary information to users. In a healthcare system, it simplifies how doctors, nurses, and hospital staff interact with patient records.
Healthcare professionals do not need to know how the database stores information internally. Instead, they access the system through user-friendly interfaces that show relevent patient data such as medical history, prescriptions, test reports, and appointment details.
Data abstraction works in three levels:

1- Physical level -
This level defines how data is actually stored in the database using storage structures. Healthcare professionals do not interact with this level.

2- Logical level -
This level describes what data is stored in the database and the relationships b/w the data. For ex, relationships between patients, doctors, medical records, & treatments.

3- View level -
This level shows only the required information to specific users. for eg:
- Doctors can see diagnosis and treatment history.
- Nurses can view medication schedules.
- Receptionists can access appointment information.

Because of data abstraction, healthcare staff can easily retrive patient records, update information, and make medical decisions

3- Explore the benefits & the challenges of implementing data abstraction in healthcare systems.

Benefits -

1. Improved Data Security : Sensitive patient information is protected because users can only access the data they are authorized to see.
2- Simplified User Interaction: Doctors and staff interact with simple interfaces rather than complex database systems.
3- Better Data Management : Data abstraction organizes large amounts of healthcare data efficiently.
4- Reduced Complexity : It hides complicated database structures from users, making systems easier to operate.
5- Improved Data Consistency : Rules and constraints ensure that medical records remain accurate and reliable.

Challenges

1. Implementation Complexity : Designing abstraction layers in healthcare databases can be technically complex.
2. System Cost : Developing and maintaining advanced healthcare information systems can be expensive.
3. Training Requirements : Hospital staff may require training to use new systems effectively.
4. Data integration issues: Integrating data from different hospital departments can sometimes be difficult.

Despite these challenges, data abstraction significantly improves the efficiency and reliability of healthcare data management.

Section - C (Simulation - based)

Q.3. Design a database for a university, incorporating entities like students, courses, and professors. Using the entity-relationship model, illustrate and explain the relationships between these entities. Include attributes for each entity to capture relevant information such as student names, course details, and professor information.

A university database can include entities such as Students, Courses, and Professors. These entities are connected through relationships.

Entities & Attributes -

1- ~~Attributes~~ Student - Attributes:
- Student_ID (PK)
- Student_Name
- Email
- Phone
- Department

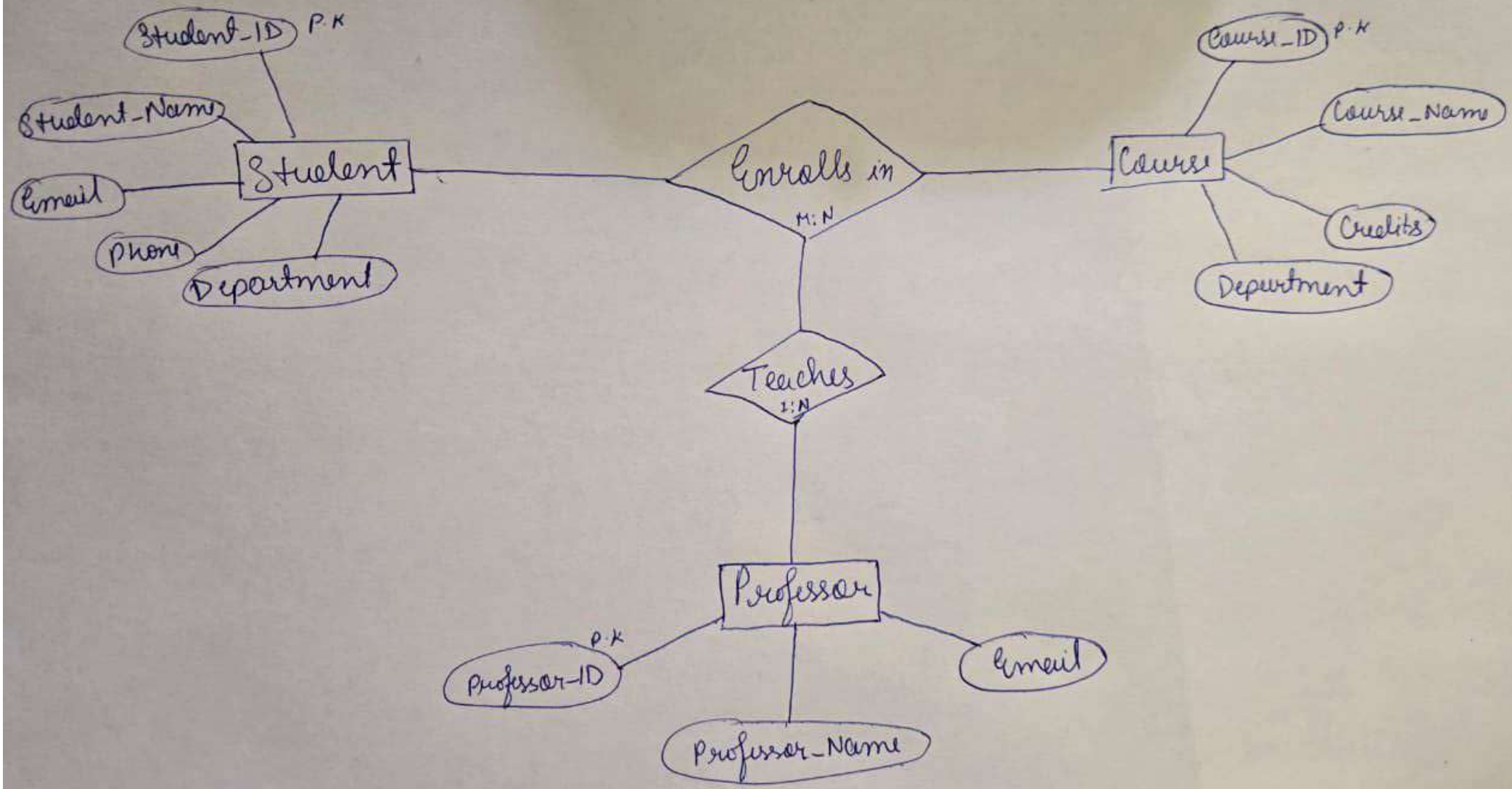
2- Course
- Course_ID (PK)
- Course_Name
- Credits
- Department

3- Professor
- Professor_ID (P.K)
- Professor_Name
- Email
- Department

Relationships -

1- Enrollment (Student - Course)
- A student can enroll in many courses
- A course can have many students.
- This is a many-to-many relationship.

2- Teaching (Professor - Course)
- A professor can teach multiple courses.
- A course is usually taught by one professor.
- This is a one-to-many relationship.


Student_ID
P.K
Student_Name
Student
Email
Phone
Department
Enrolls in
M:N
Course
Course_ID
P.K
Course_Name
Credits
Department
Teaches
1:N
Professor
P.K
Professor-ID
Email
Professor_Name